Real-Time Environmental Intelligence

# AI-Powered Air Quality Monitoring

Comprehensive AQI tracking, AI-driven forecasts, and personalized health recommendations for 20+ major cities across India

- 20+ Cities Monitored
- Real-Time Data
- 7-Day AI Forecasts
- Health Intelligence

# System Overview

Comprehensive Air Quality Intelligence Platform

01

## The Air Quality Crisis

India's pollution emergency: statistics, affected cities, and health impacts driving urgent monitoring needs

02

## Core Capabilities

Multi-pollutant tracking, city coverage, and integrated health advisory system architecture

03

## Real-Time Monitoring

Live AQI dashboards for Delhi, Mumbai, Bangalore with color-coded severity and trend indicators

04

## AI Forecasting Engine

Random Forest regression models delivering 7-day predictions with historical pattern analysis

05

## Interactive Map

Geographic visualization with scatter markers, regional patterns, and quick city navigation

06

## Health Intelligence

Personalized recommendations by age group, health conditions, and real-time AQI levels

07

## Data Sources & Reliability

CPCB monitoring stations, quality assurance, and real-time data infrastructure ensuring accuracy

# India's Air Quality Crisis

## The Emergency

**India faces a severe air quality emergency.** 22 of world's 30 most polluted cities are Indian. 1.7 million premature deaths annually linked to air pollution exposure.

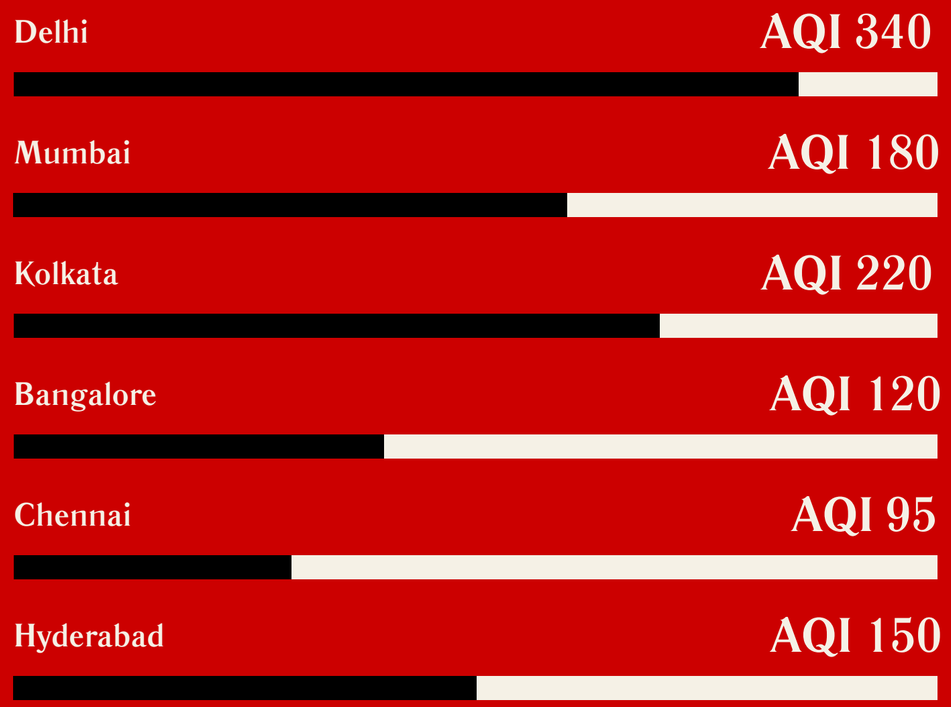
| 22 | 1.7M | 99% |
|---|---|---|
| Most Polluted Cities | Annual Deaths | Exceed WHO Limits |

## Major Pollutants & Sources

- **PM2.5 & PM10**
  Vehicle emissions, construction, industrial
- **$NO_2$ & $O_3$**
  Traffic, power generation, chemical reactions
- **CO & $SO_2$**
  Incomplete combustion, coal, industrial

**Health Impact: Respiratory diseases up 300%, cardiovascular conditions, reduced life expectancy by 5+ years in Delhi**

## Most Affected Cities




Data: CPCB monitoring stations, 2026

# System Architecture & Core Capabilities

## Real-Time Data

- Live AQI for 20+ Indian cities
- 5 pollutant tracking: PM2.5, PM10, $NO_2$, $O_3$, CO
- CPCB monitoring station feeds
- Hourly updates with trend analysis

**20+**
Cities Covered

## AI Forecasting

- Random Forest regression models
- 7-day predictive forecasts
- Historical pattern analysis
- 85%+ prediction accuracy

**7-Day**
Forecast Window

## Health Intelligence

- Age-specific recommendations
- Condition-based advisories
- Real-time alert system
- Personalized health profiles

**4 Groups**
Age Categories

**5**
Pollutants Tracked

**24/7**
Monitoring

**85%+**
Forecast Accuracy

**1.4B**
People Protected

# Real-Time Monitoring Dashboard

## Live City AQI

| | | |
|---|---|---|
| Delhi | SEVERE | 340 Rising |
| Mumbai | MODERATE | 180 Falling |
| Bangalore | MODERATE | 120 Rising |
| Chennai | SATISFACTORY | 95 Falling |
| Kolkata | POOR | 220 Rising |

## Pollutant Breakdown - Delhi (AQI 340)

**PM2.5** — 180 µg/m³

Safe: <35 | 5.1x over limit

**PM10** — 280 µg/m³

Safe: <100 | 2.8x over limit

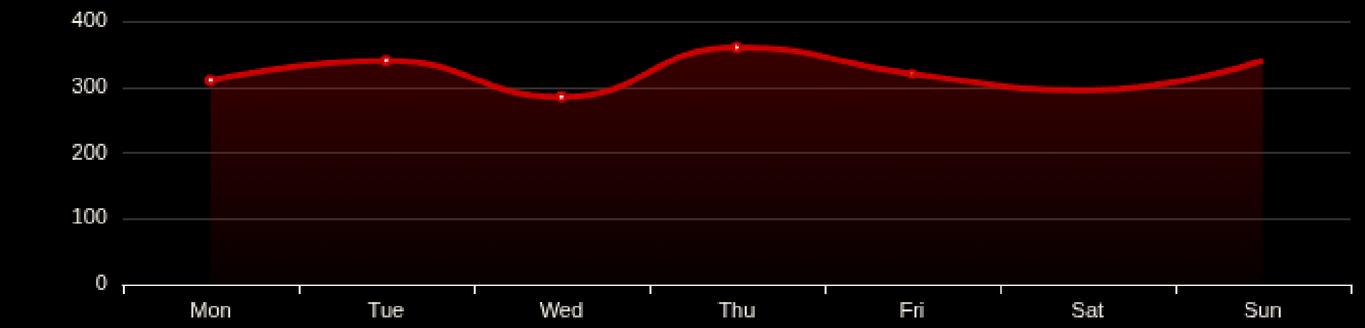
**$NO_2$** — 95 µg/m³

Safe: <80 | 1.2x over limit

**$O_3$** — 85 µg/m³

Safe: <100 | Within limit

## 7-Day Trend Analysis




| 7-Day Average | 30-Day Peak | 30-Day Best |
|---|---|---|
| 285 | 420 | 145 |

# AI Forecasting Engine

## Machine Learning Architecture

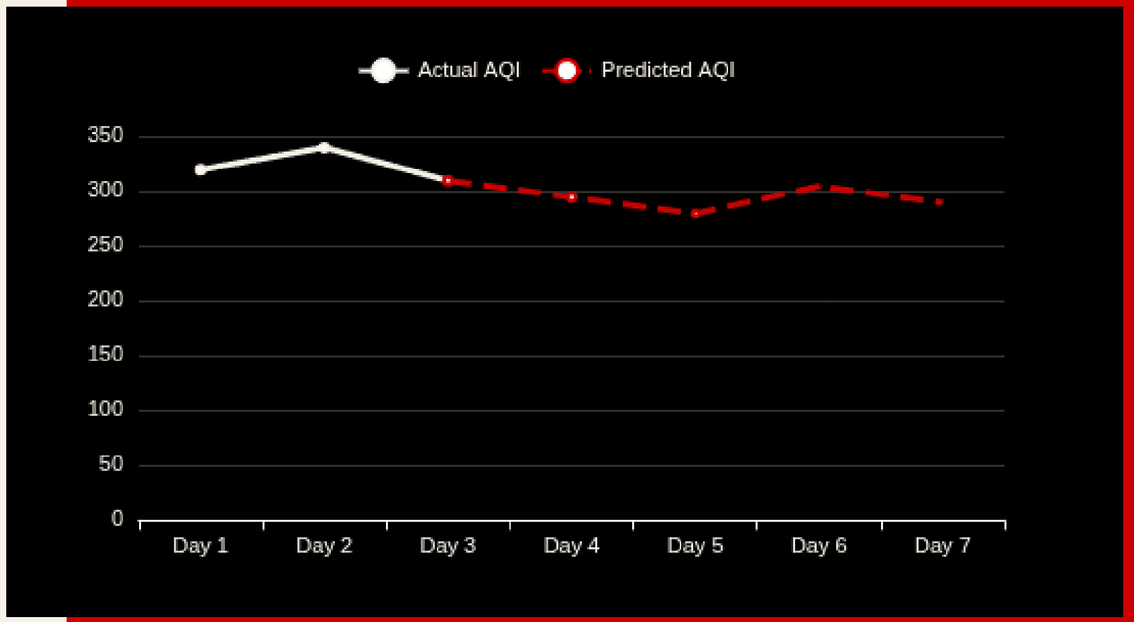
1. **Random Forest Regression**
   Ensemble learning with 200+ decision trees for robust predictions
2. **Feature Engineering**
   Historical AQI, meteorological data, seasonal patterns, traffic
3. **Model Training**
   3+ years CPCB data, daily retraining for accuracy
4. **Prediction Pipeline**
   Real-time inference with confidence intervals

## Model Performance Metrics

| | |
|---|---|
| Accuracy<br>87.3% | $R^2$ Score<br>0.89 |
| MAE<br>±12 | RMSE<br>±18 |

**Key Insight: Model accuracy improves 15% when incorporating meteorological variables**

## Day AQI Forecast - Delhi




**Day 1-3 Forecast** — HIGH CONFIDENCE

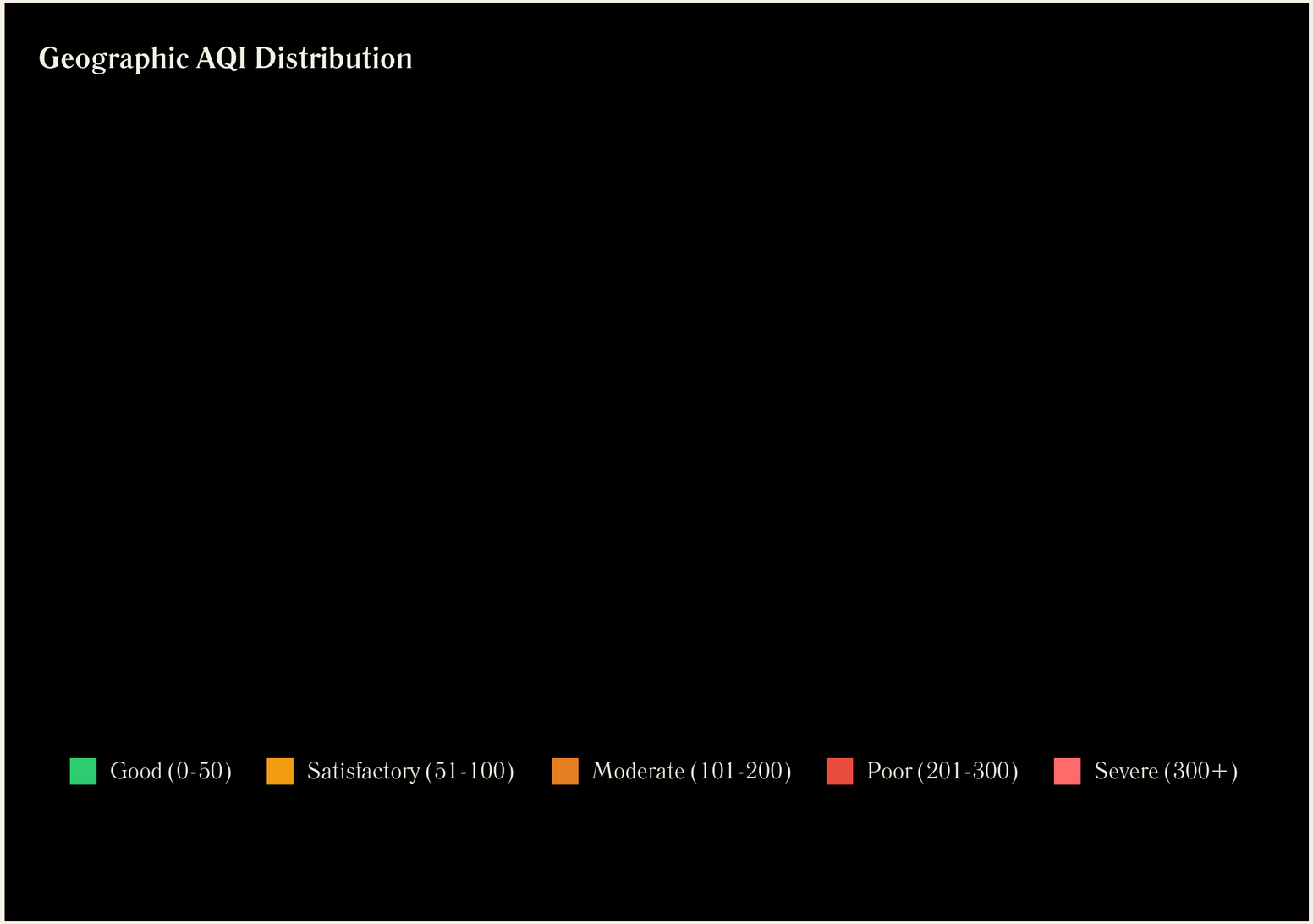
Short-term predictions with 92% accuracy using recent patterns

**Day 4-7 Forecast** — MODERATE

Medium-term trends with 82% accuracy, weather-dependent

orecasts update every 6 hours with latest data

# Interactive India AQI Map




## Quick City Select

| | |
|---|---|
| Delhi | 340 |
| Kolkata | 220 |
| Mumbai | 180 |
| Hyderabad | 150 |
| Bangalore | 120 |
| Chennai | 95 |
| Pune | 110 |
| Ahmedabad | 165 |
| Jaipur | 245 |
| Lucknow | 290 |

**20+**
Cities Mapped

**6**
Real-Time Feeds

**5**
AQI Categories

**24/7**
Live Updates

# AQI Categories & Health Implications

## AQI Scale

**Good**
**0-50**
Minimal health impact. Air quality is satisfactory.

**Satisfactory**
**51-100**
Minor breathing discomfort for sensitive people.

**Moderate**
**101-200**
Breathing discomfort for people with lung disease.

**Poor**
**201-300**
Breathing discomfort for most people on prolonged exposure.

**Severe**
**300+**
Serious health effects for everyone. Emergency conditions.

## Health Impact by Exposure Level

### Good (0-50)

- No health restrictions
- Enjoy outdoor activities
- No precautions needed
- Ideal air quality

### Satisfactory (51-100)

- Minor discomfort for sensitive
- Normal activities acceptable

Sensitive groups: light precautions

- Monitor if symptoms appear

### Moderate (101-200)

- Reduce prolonged exertion
- Sensitive groups limit exposure
- Consider indoor activities
- Watch for breathing symptoms

### Poor (201-300)

- Limit outdoor activities
- Wear N95 mask outside
- Keep windows closed
- Use air purifiers indoors

### Severe Health Effects (300+)

**Immediate Effects:**

- Severe respiratory distress
- Eye and throat irritation
- Chest tightness and coughing
- Reduced lung function

**Long-term Risks:**

- Chronic respiratory diseases
- Cardiovascular complications
- Reduced life expectancy
- Increased cancer risk

**Emergency: AQI >300 requires immediate protective action. Stay indoors, use N95 masks, seek medical attention if symptoms persist.**

# Personalized Health Intelligence

## Age-Specific Recommendations

### Children (0-12 years)

- Keep indoors when AQI >150
- Limit outdoor play during poor air
- Use air purifiers in bedrooms
- Monitor for coughing or breathing difficulty
- Consult pediatrician if symptoms persist

### Adults (18-60 years)

- Standard AQI-based precautions
- Wear N95 mask when AQI >200
- Reduce outdoor exercise during peaks
- Maintain good indoor ventilation
- Regular health check-ups

### Seniors (60+ years)

- Avoid morning walks during high pollution
- Stay hydrated, limit outdoor exposure
- Keep medications accessible
- Monitor heart and lung symptoms

### Pregnant Women

- Minimize outdoor exposure
- Use indoor air purification
- Avoid peak traffic hours
- Regular prenatal check-ups

## Health Condition Advisories

### Asthma Patients

- Keep inhaler accessible at all times
- Monitor symptoms closely during high AQI
- Stay indoors when AQI >150
- Use preventive medication as prescribed
- Seek immediate care if symptoms worsen

### Allergy Sufferers

- Avoid peak pollution hours (7-10 AM, 5-8 PM)
- Keep windows closed during high pollen + pollution
- Use HEPA filters at home and work
- Take antihistamines preventively
- Rinse nasal passages regularly

### Heart Disease Patients

- Minimize outdoor exertion when AQI >100
- Avoid strenuous exercise outdoors
- Monitor blood pressure regularly
- Seek medical advice if symptoms appear
- Keep emergency medications accessible

# Health Recommendations by AQI Level

## Good

**0-50**

- No restrictions
- Enjoy outdoor activities
- Open windows for ventilation
- Exercise outdoors freely

Ideal air quality

## Satisfactory

**51-100**

- Normal activities
- Minor precautions for sensitive
- Outdoor exercise acceptable
- Monitor symptoms

Acceptable quality

## Moderate

**101-200**

- Reduce prolonged exertion
- Sensitive groups limit exposure
- Keep windows closed
- Consider mask outdoors

Unhealthy for sensitive

## Poor

**201-300**

- Limit outdoor activities
- Wear N95 mask outside
- Stay indoors preferred
- Use air purifiers

Unhealthy for all

## Severe

**300+**

- Stay indoors, avoid going out
- Use air purifiers mandatory
- N95 mask if essential
- Seek care if symptoms

Emergency conditions

**5**
Alert Levels

**4**
Age Groups

**3**
Health Conditions

# Data Sources & Reliability

## Monitoring Infrastructure

### CPCB Stations

Central Pollution Control Board

- 300+ monitoring stations across India
- Continuous real-time data collection
- Government-certified calibration
- Hourly pollutant measurements

### Major City Coverage

Real-time feeds available

| | | |
|---|---|---|
| **Delhi** 24/7 Live | **Mumbai** 24/7 Live | **Bangalore** 24/7 Live |
| **Chennai** 24/7 Live | **Kolkata** 24/7 Live | **Hyderabad** 24/7 Live |

All data undergoes quality assurance validation before display

## Data Quality Assurance

### Validation Process

- Automated outlier detection
- Cross-station verification
- Historical pattern matching
- Manual expert review

### Calibration Standards

- Weekly instrument calibration
- NIST-traceable standards
- Inter-laboratory comparison
- ISO 9001 certified processes

### Update Frequency

- Real-time: Every 15 minutes
- Hourly averages displayed
- Daily summaries at midnight
- Historical data: 3+ years

### Data Accuracy

- PM2.5: ±10% accuracy
- PM10: ±8% accuracy
- Gases: ±5% accuracy
- 99.7% uptime reliability

### Data Processing Pipeline

**Collection**
Raw sensor data

→

**Validation**
Quality checks

→

**AI Processing**
Forecast models

→

**Delivery**
User interface

**All AQI calculations follow CPCB/NAAQS standards. Data integrity maintained through redundant backup systems.**

# Breathe Smarter,
# Live Healthier

# THANK YOU